Analysis of food-allergic and anaphylactic events in the National Electronic Injury Surveillance System.

BACKGROUND: The National Electronic Injury Surveillance System (NEISS) captures a nationally representative probability sample from hospital emergency departments (EDs) in the United States. OBJECTIVE: Emergency department data from NEISS were analyzed to assess the magnitude and severity of adverse events attributable to food allergies. METHODS: Emergency department events describing food-related allergic symptomatology were identified from 34 participating EDs from August 1 to September 30, 2003. RESULTS: Extrapolation of NEISS event data predicts a total of 20,821 hospital ED visits, 2333 visits for anaphylaxis, and 520 hospitalizations caused by food allergy in the United States during the 2-month study period. The median age was 26 years; 24% of visits involved children < or =5 years old. Shellfish was the most frequently implicated food in persons > or =6 years old, whereas children < or =5 years old experienced more events from eggs, fruit, peanuts, and tree nuts. There were no reported deaths. Review of medical records found that only 19% of patients received epinephrine, and, using criteria established by a 2005 anaphylaxis symposium, 57% of likely anaphylactic events did not have an ED diagnosis of anaphylaxis. CONCLUSION: Analysis of NEISS data may be a useful tool for assessing the magnitude and severity of food-allergic events. A criteria-based review of medical records suggests underdiagnosis of anaphylactic events in EDs.